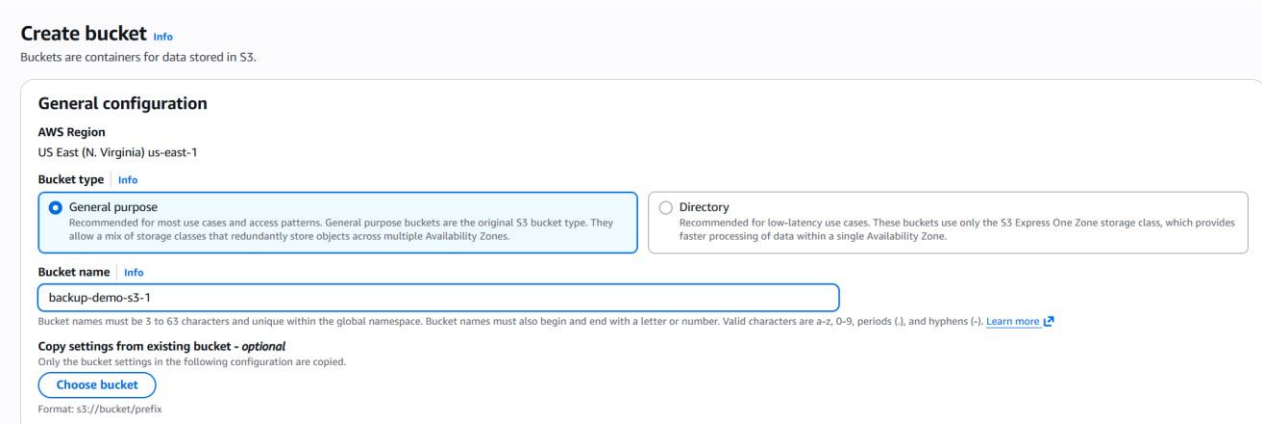
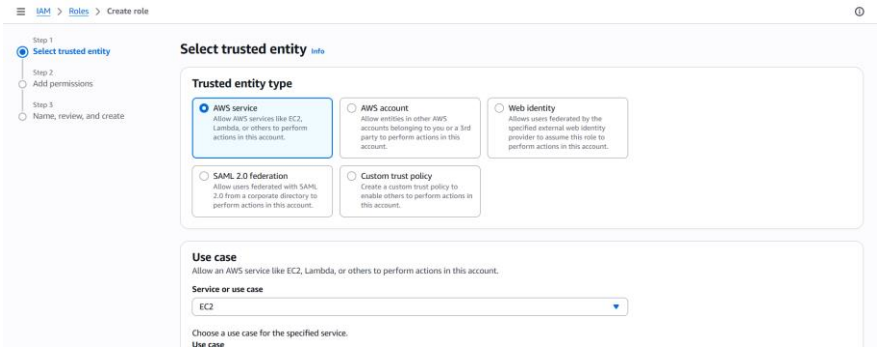
# EC2 Automatic Backup System to S3 (Linux)

I implemented an automated backup system where a Linux server running on Amazon EC2 securely uploads important files to Amazon S3. Instead of storing access keys on the server, I used an IAM Role through AWS Identity and Access Management to provide secure access. The backup process was automated using a shell script and scheduled with a cron job, ensuring regular and reliable data protection without manual intervention.

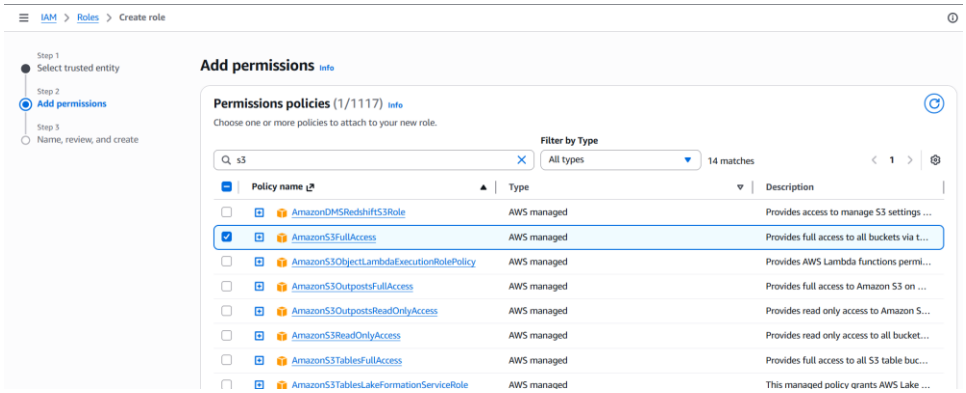
**Step 1 :** create an s3 bucket



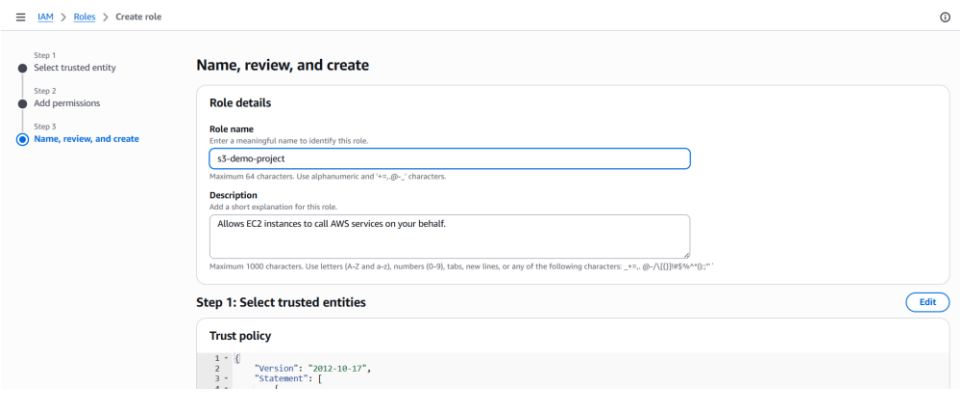

- Ater creating the bucket
- Create an iam role



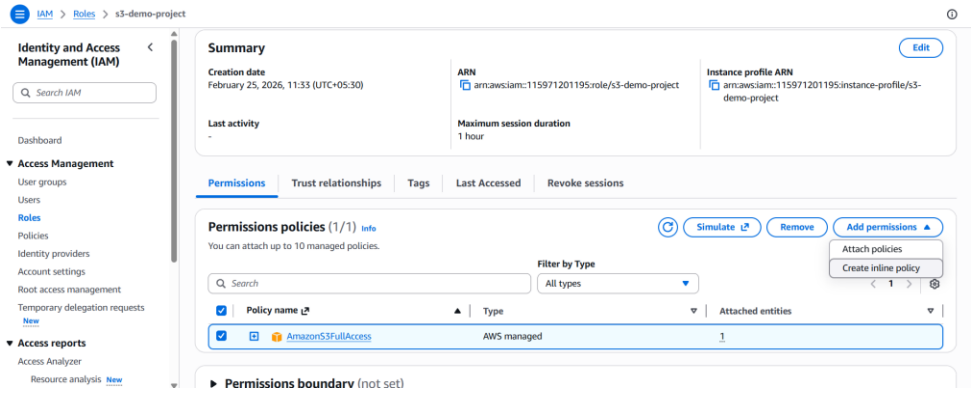
- On that  choose aws
- After that choose s3 full access on the permission



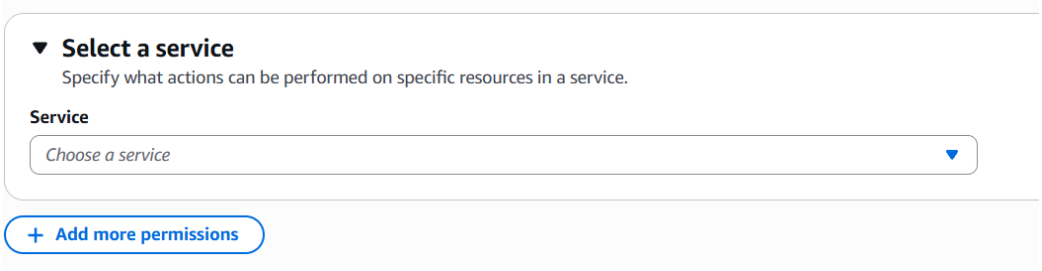

- After that choose the role name




- After that go the iam  role choose the iam role you have created



- After selecting the i am role click add permission
- On the permission choose create inline policy



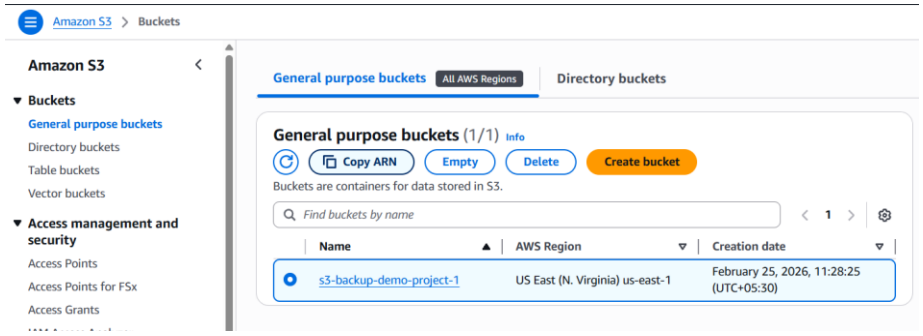

- Choose the serive you want to use
- After that on the right side corner you can see json

Attach given json formate

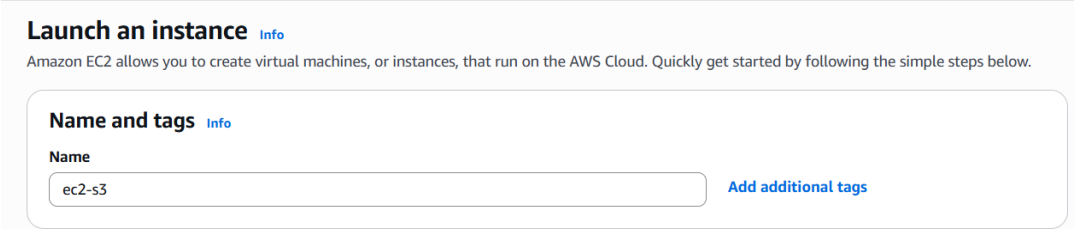
```
{
 "Version": "2012-10-17",
 "Statement": [
{
    "Effect": "Allow",
    "Action": [
    "s3:PutObject",
    "s3:GetObject",
    "s3:ListBucket"
    ],
    "Resource": [
    "arn:aws:s3:::myec2backup12",
    "arn:aws:s3:::myec2backup12/*"
```

```
    ]
}
 ]
}
```

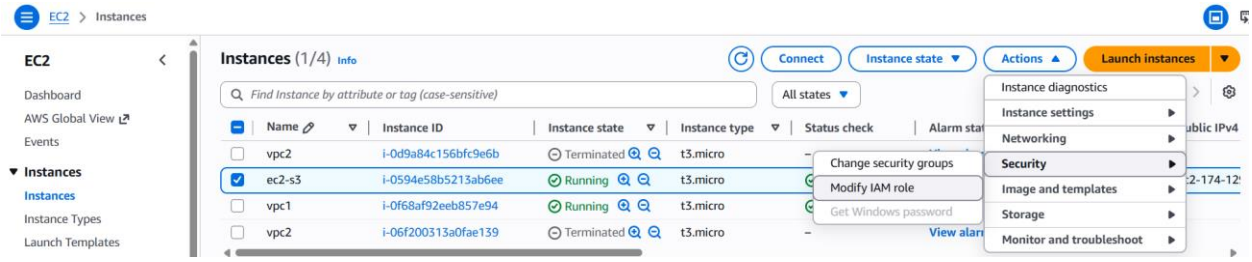
- Change the bucket name insert the bucket namr you have created



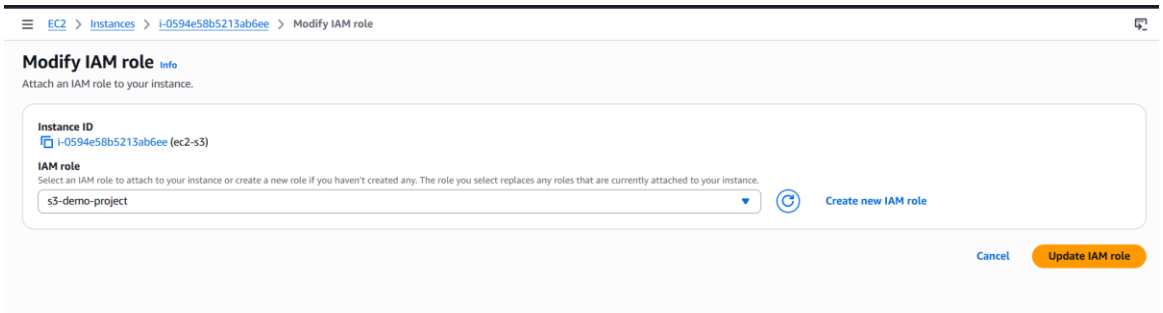

- Next attach and save it.
- Create a linux sever




- After launching the instance




- On that choose your ec2 click actions
- On that actions choose security
- On that security choose modify iam role




- On that choose your role
- After that connect your linux server

```
PS C:\WINDOWS\System32> cd ~
PS C:\Users\selva> cd .\Downloads\
PS C:\Users\selva\Downloads> ssh -i "backup.pem" ec2-user@ec2-52-72-170-95.compute-1.amazonaws.com
```

- After that

```
        _/m/'
[ec2-user@ip-172-31-34-24 ~]$ sudo vi /usr/local/bin/backup_to_s3.sh
```

- In this we first change into root user
- And then we have created a vim file
- After that we have given a path
- After that we said that move the backup to s3

sudo vi /usr/local/bin/backup_to_s3.sh

- After the vim file created

Then file open in your terminal and put the command . save and exit

```
#!/bin/bash

BUCKET_NAME="myec2backup12"

SOURCE_DIR="/home/ec2-user/important_data"
```

```
TEMP_DIR="/tmp/backup_temp"

DATE=$(date +%Y-%m-%d)

TIME=$(date +%H-%M-%S)

BACKUP_FILE="backup-$DATE-$TIME.tar.gz"

echo "Starting Backup: $DATE $TIME"

mkdir -p $TEMP_DIR

echo "Compressing files..."

tar -czf $TEMP_DIR/$BACKUP_FILE $SOURCE_DIR

echo "Uploading to S3..."

aws s3 cp $TEMP_DIR/$BACKUP_FILE s3://$BUCKET_NAME/

if [ $? -eq 0 ]; then

echo "Upload Successful!"

rm $TEMP_DIR/$BACKUP_FILE

else

echo "Upload Failed! Check logs."

fi
```

Then Save and exit :wq!

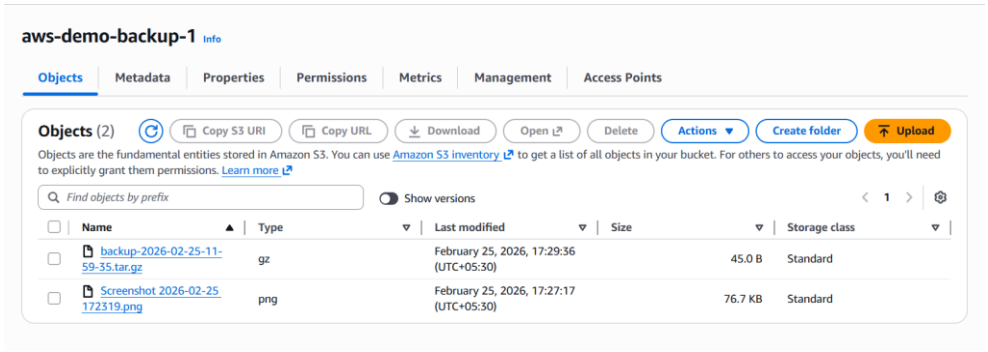
**sudo chmod +x /usr/local/bin/backup_to_s3.sh**

**/usr/local/bin/backup_to_s3.sh**

- After that

```
[ec2-user@ip-172-31-34-24 ~]$ sudo chmod +x /usr/local/bin/backup_to_s3.sh
[ec2-user@ip-172-31-34-24 ~]$ /usr/local/bin/backup_to_s3.sh
```

  - On this we change the directory and execute the path of the bucket
  - After the test the path of the bucket
  - It will show the back up file on the s3 file




- Okay now the back has been created now we want to cretae the automatic backup
- For that wee need cronie

```
[ec2-user@ip-172-31-19-125 ~]$ sudo yum install cronie -y
```

- Install cronie
- After that

**Sudo yum install cronie -y**

**sudo systemctl enable crond**
**sudo systemctl start crond**

**sudo systemctl status crond**

```
[ec2-user@ip-172-31-19-125 ~]$ sudo systemctl enable crond
[ec2-user@ip-172-31-19-125 ~]$ sudo systemctl start crond
[ec2-user@ip-172-31-19-125 ~]$ sudo systemctl status crond
● crond.service - Command Scheduler
```

- In this we first enable the cronie
- Second start the cronie
- Check the status
- **Crontab -e**

```
[ec2-user@ip-172-31-19-125 ~]$ crontab -e
no crontab for ec2-user - using an empty one
crontab: installing new crontab
```

- It open a dialog box
- Insert this

**0 2 * * * /usr/local/bin/ec2-s3-backup.sh >> /var/log/ec2-backup.log 2>&1**

```
┌──────── Minute (0 - 59)
│ ┌────── Hour (0 - 23)
│ │ ┌──── Day of Month (1 - 31)
│ │ │ ┌── Month (1 - 12)
│ │ │ │ ┌ Day of Week (0 - 7)
│ │ │ │ │
```

0 2 * * *

- This work on this basic
- We can change the timing in this
- It atomatiicaly backup